UNIVERSITY OF RWANDA

COLLEGE OF SCIENCE AND TECHNOLOGY SCHOOL OF ICT

COMPUTER AND SOFTWARE ENGINEERING DEPARTMENT

NYARUGENGE CAMPUS

MODULE CODE/TITLE: WEB DESIGN

AI FLASHCARDS MAKER SYSTEM WEB APP, Case of Primary, Secondary and University Students Studying Guide

225033151

Year of study: 1

Date: July 18 2026

Project Technical Report- Web Technology TECHNICAL ASSESSMENT 2026

Table of contents

UNIVERSITY OF RWANDA 1
COLLEGE OF SCIENCE AND TECHNOLOGY SCHOOL OF ICT 1
COMPUTER AND SOFTWARE ENGINEERING DEPARTMENT NYARUGENGE CAMPUS 1
Project Technical Report- Web Technology TECHNICAL ASSESSMENT 2026....1
Table of contents 1
Table of Figures....1
List of Abbreviations 1
Chapter 1. Introduction 2
1.1 Historical Background of the Case Study....2
1.2 Problem Statement....2
1.3 Project Objectives....3
General Objective 3
Specific Objectives 3
1.4 Expected Outcomes....3
1.5 Project Rationale (Importance to the Community) 3
Chapter 2. System Analysis and Design 3
2.1 System Analysis....3
2.1.1 Intended Users of the Project....3
Administrator....3
Teachers....4
Students....4
2.1.2 Intended System Partners 4
Artificial Intelligence API Provider 4
Payment Gateway Provider....4
Email Service Provider 4
2.2 System Design....4
2.2.1 High-Level Architecture 4
2.2.2 UI Design and Flowcharts 4
Guest Module 5
Student Module....5
Teacher Module....5
Administrator Module 6
Overall Navigation Flow....7
2.2.3 Database Design with ERD and Relationships 7
Relationships....8

Chapter 3. Implementation 8
3.1 Introduction 9
3.2 Screenshots and Discussion 9
3.2.1 Landing Page 9
3.2.2 Login Page 9
3.2.3 Registration Page 10
3.2.4 Student Dashboard 10
3.2.5 Teacher Dashboard 11
3.2.6 Class Management Page 11
3.2.7 Flashcard Generation Page 11
3.2.8 Subscription Page 11
3.2.9 Admin Dashboard 11
Chapter 4. Conclusion and Recommendation 12
Conclusion 12
Recommendations 12
To Lecturers 12
To the School 12
To Educational Institutions 12
To Users 12
To Future Researchers 12
Appendix 12

List of Abbreviations

AI – Artificial Intelligence
API – Application Programming Interface
ERD – Entity Relationship Diagram
UI – User Interface
UX – User Experience
DBMS – Database Management System
CRUD – Create, Read, Update, Delete
ICT – Information and Communication Technology
PAY – Payment Processing System

Chapter 1. Introduction

1.1 Historical Background of the Case Study

Traditional learning methods often rely on handwritten notes and repetitive reading, which can make memorization difficult and time-consuming. Flashcards have long been used as an effective study technique because they encourage active recall and spaced repetition.

With the advancement of Artificial Intelligence, educational technologies have become more capable of automatically generating learning materials from user input. AI can reduce the time required to prepare study resources and improve learning efficiency.

The AI Flashcard Maker system is designed to help teachers and students generate, organize, and share flashcards automatically using Artificial Intelligence. The system also supports classroom management, subscription services, and collaborative learning.

1.2 Problem Statement

Many teachers and students spend significant time manually creating study materials. Existing flashcard systems often lack intelligent content generation and classroom collaboration features.

The major problems identified are:

- Manual creation of flashcards is time-consuming.

- Difficulty sharing learning materials among students.
- Lack of centralized classroom management.
- Limited integration of Artificial Intelligence in educational platforms.
- Existing solutions often require paid subscriptions for all users.

1.3 Project Objectives

General Objective

To develop a web-based AI Flashcard Maker system that assists teachers and students in creating and managing flashcards efficiently.

Specific Objectives

1. To provide AI-powered flashcard generation.
2. To allow teachers to create and manage classes.
3. To allow teachers to share flashcards with specific classes.
4. To enable students to create their own flashcards.
5. To implement authentication and authorization mechanisms.
6. To implement subscription and payment management.
7. To provide an administration panel for system management.

1.4 Expected Outcomes

The project is expected to:

- Reduce the time required to create study materials.
- Improve collaboration between teachers and students.
- Provide intelligent flashcard generation using AI.
- Improve learning effectiveness through digital study tools.
- Provide a scalable educational platform.

1.5 Project Rationale (Importance to the Community)

The system contributes to digital education by simplifying learning material preparation and improving accessibility. Students can study more efficiently, while teachers can manage educational resources more effectively.

The system also promotes technology adoption within educational institutions and supports modern teaching methodologies.

Chapter 2. System Analysis and Design

2.1 System Analysis

2.1.1 Intended Users of the Project

Administrator

Responsibilities:

- Manage users.
- Manage subscription plans.
- Manage payments.
- Monitor system activities.

Teachers

Responsibilities:

- Create and manage classes.
- Generate flashcards using AI.
- Share flashcards with students.
- Manage subscriptions.

Students

Responsibilities:

- Join classes.
- Access shared flashcards.
- Generate personal flashcards.
- Study and manage flashcard collections.

2.1.2 Intended System Partners

Artificial Intelligence API Provider

Provides AI services used to generate flashcards automatically.

Payment Gateway Provider

Processes subscription payments securely.

Email Service Provider

Handles account verification and notifications.

2.2 System Design

2.2.1 High-Level Architecture

Description

The system follows a client-server architecture.

The frontend communicates with backend APIs. The backend handles authentication, business logic, AI integration, subscriptions, and database operations.

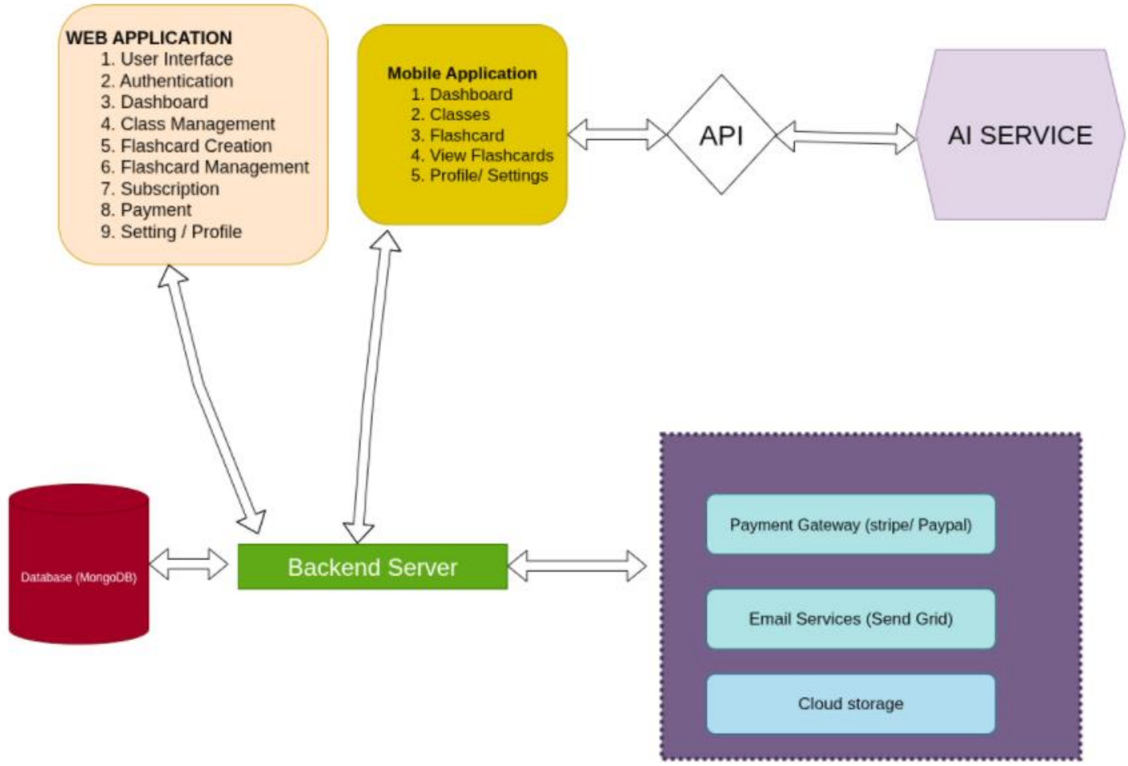


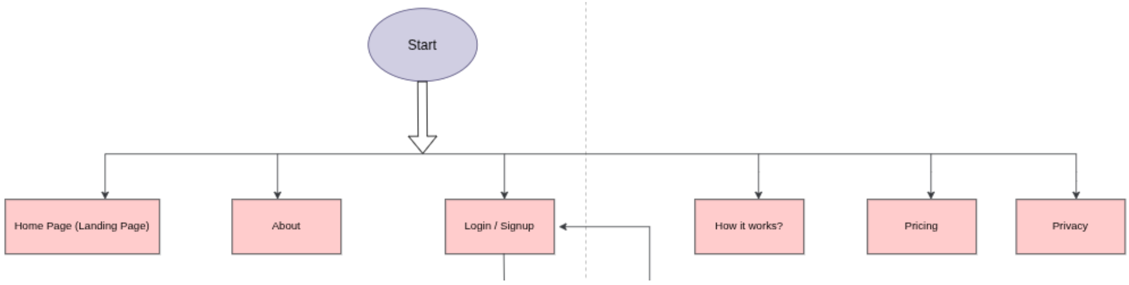
Figure 2.1: High-Level System Architecture Diagram

2.2.2 UI Design and Flowcharts

Guest Module

Pages:

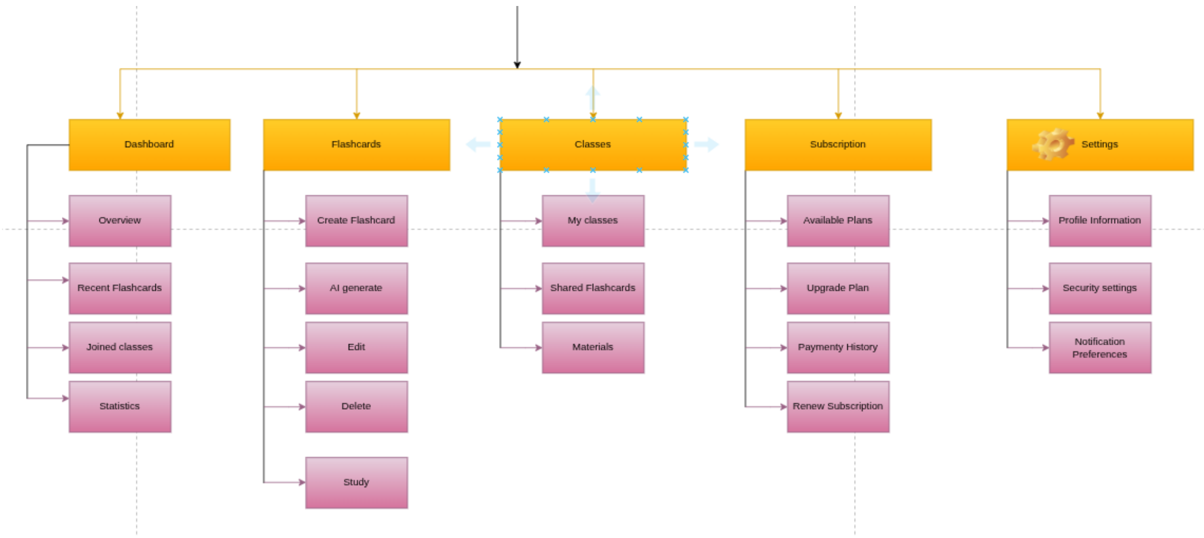
- Landing Page
- Login Page
- About Page
- How It Works? Page
- Privacy Page
- Registration Page
- Pricing Page



Student Module

Pages:

- Dashboard
- Flashcard Management
- Classes
- Subscription
- Settings/Profile



Teacher Module

Pages:

- Dashboard
- Class Management
- Flashcard Generation
- Flashcard Sharing
- Subscription
- Settings/Profile

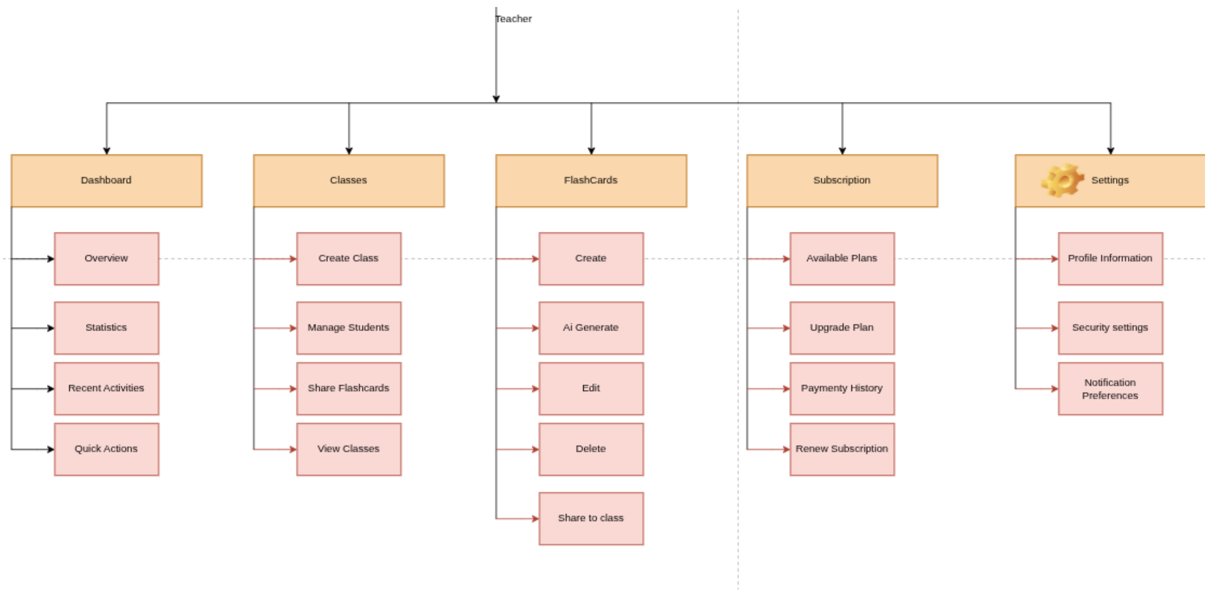


Figure 2.3: Teacher module diagram

Administrator Module

Pages:

- Dashboard
- User Management
- Subscription Plans
- Payment Management
- System Settings

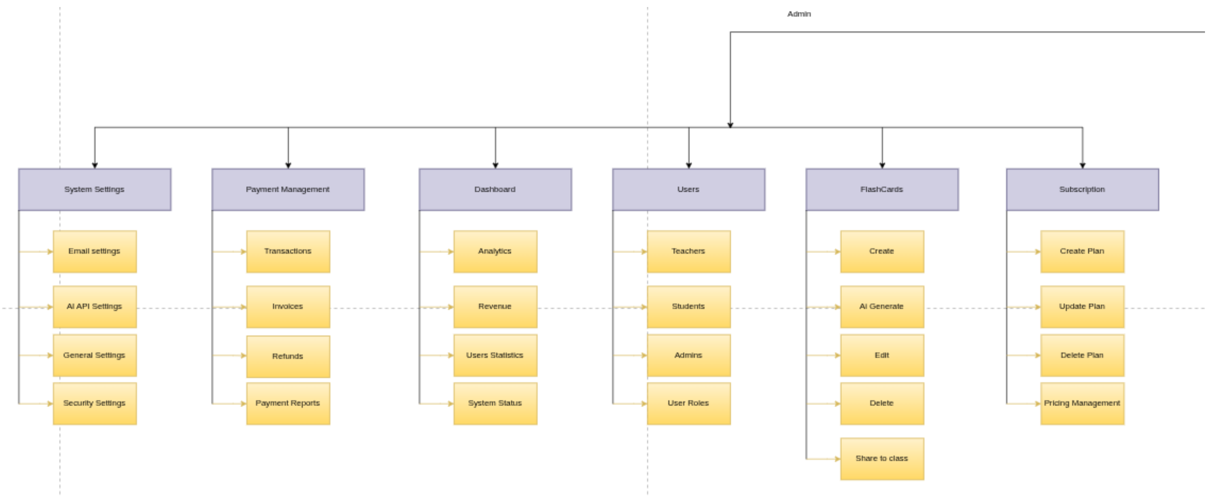


Figure 2.3 : Administrator module diagram

Overall Navigation Flow

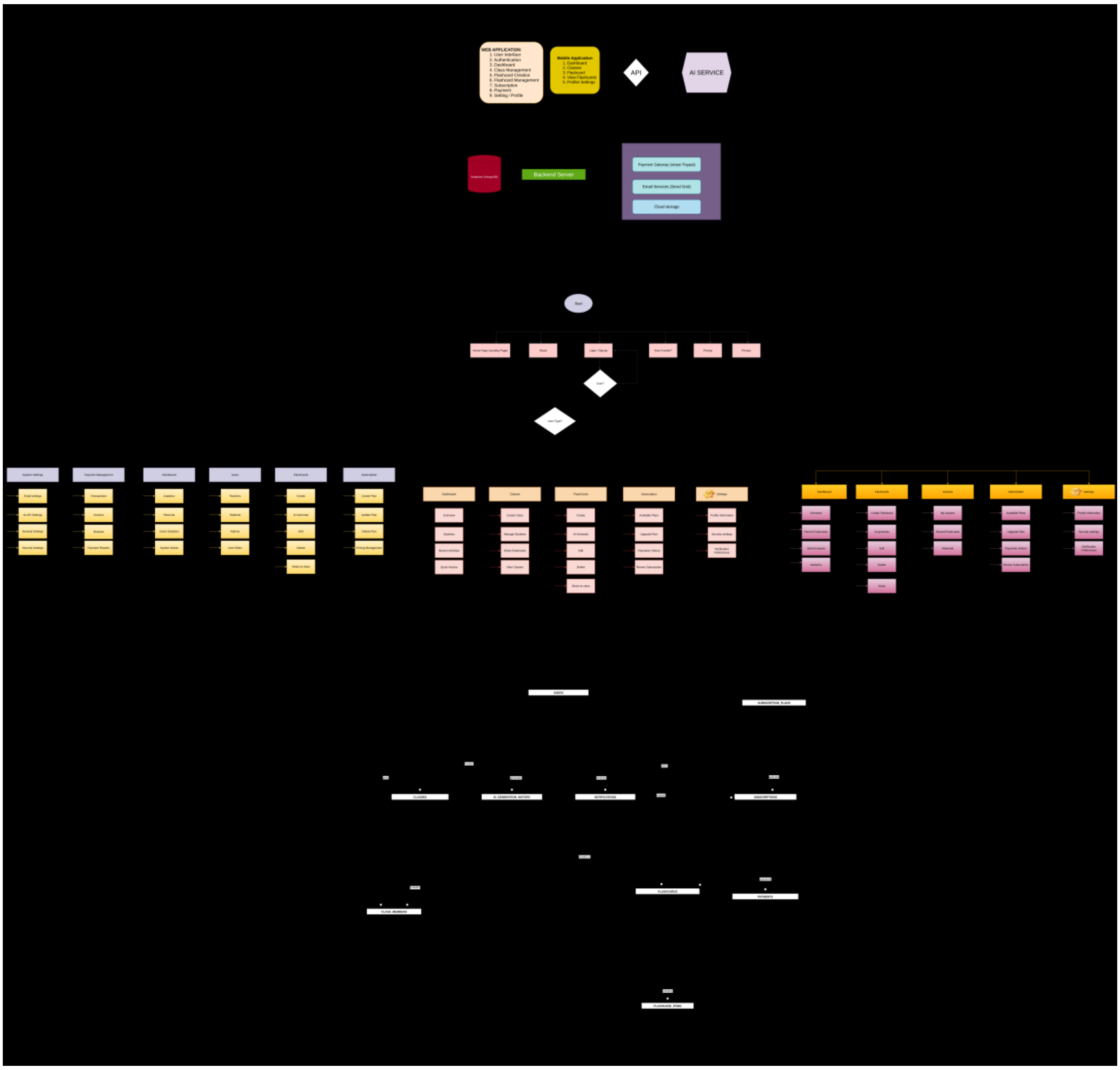


Figure 2.4: Overall system diagram

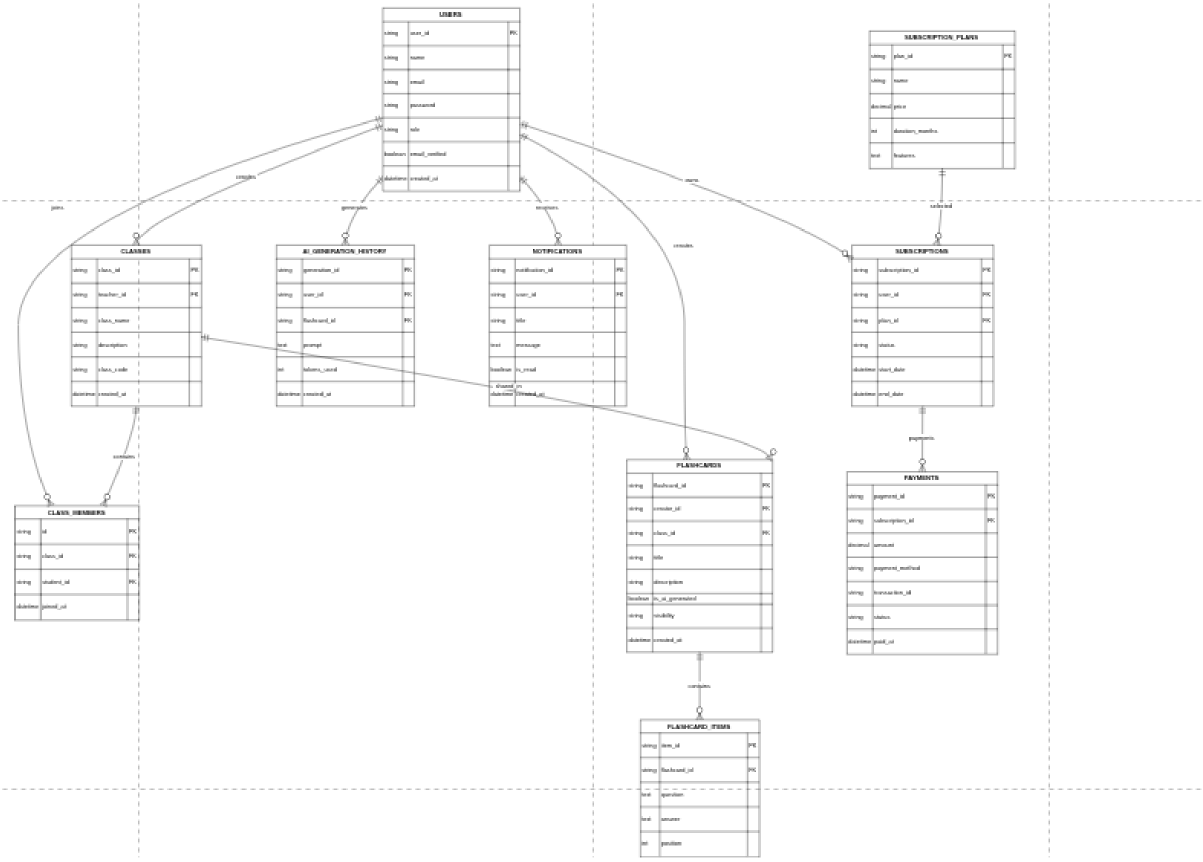
2.2.3 Database Design with ERD and Relationships

The database consists of the following entities:

- Users
- Classes
- Class Members
- Flashcards
- Flashcard Items
- Subscription Plans
- Subscriptions
- Payments
- Notifications
- AI Generation History

Relationships

1. One teacher can create many classes.
2. One class can contain many students.
3. One user can create many flashcards.
4. One flashcard can contain many flashcard items.
5. One user can own many subscriptions.
6. One subscription can have many payments.



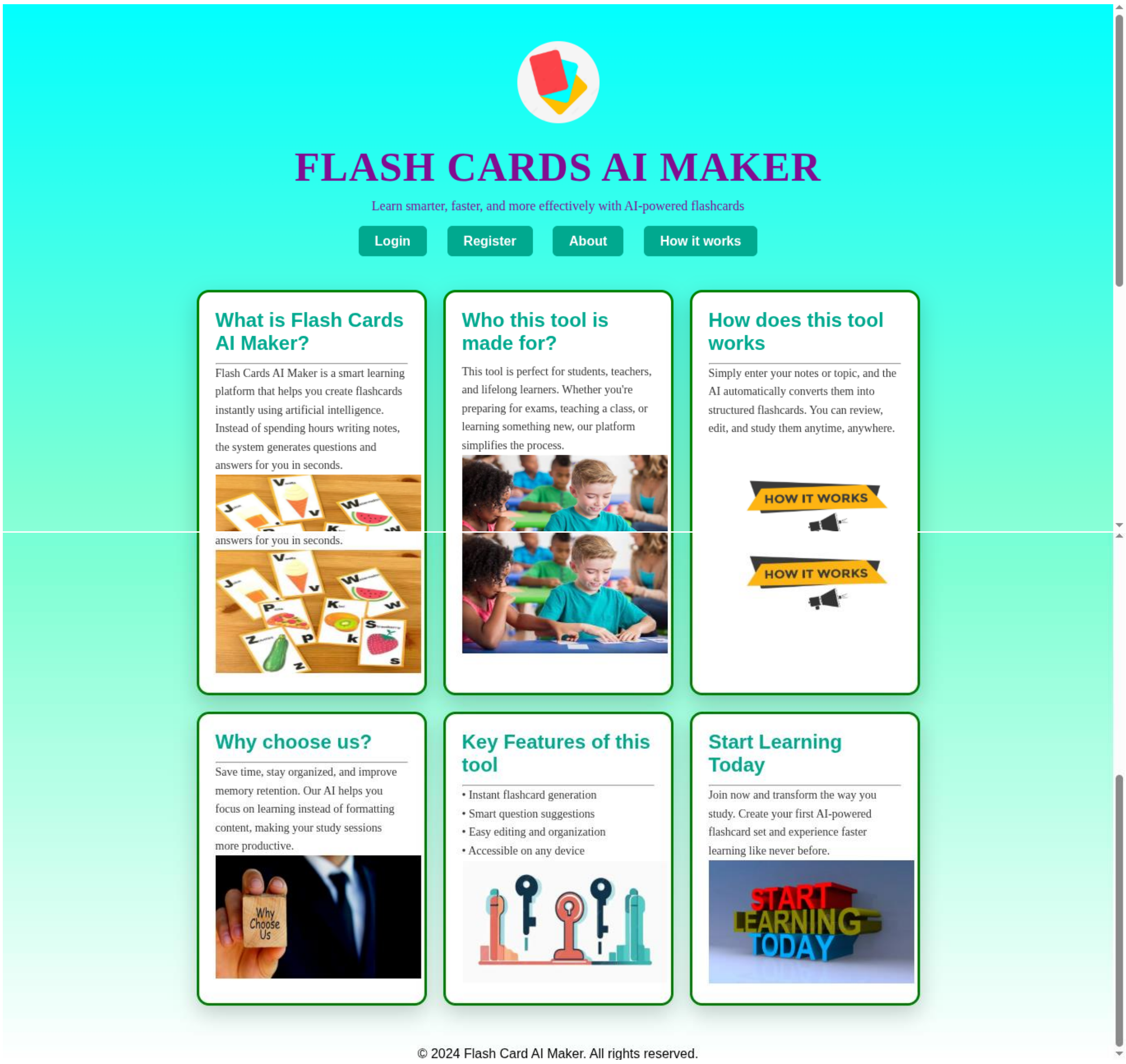
Chapter 3. Implementation

3.1 Introduction

This chapter presents the implementation of the AI Flashcard Maker system and provides screenshots of the developed modules.

3.2 Screenshots and Discussion

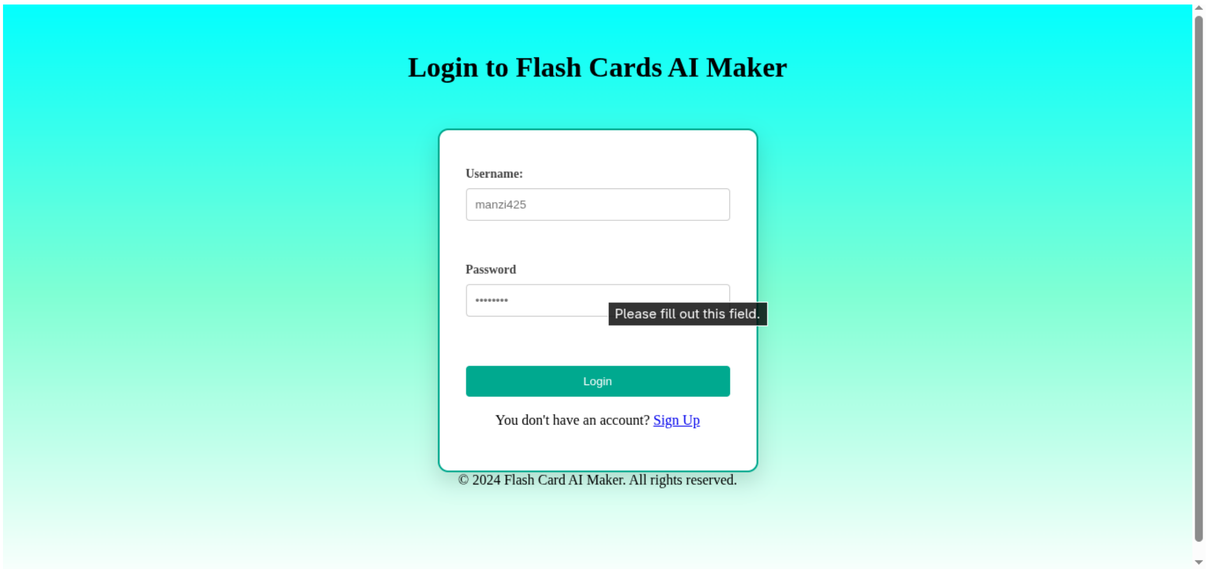
3.2.1 Landing Page



Discussion:

The landing page introduces the system features, pricing information, and authentication options.

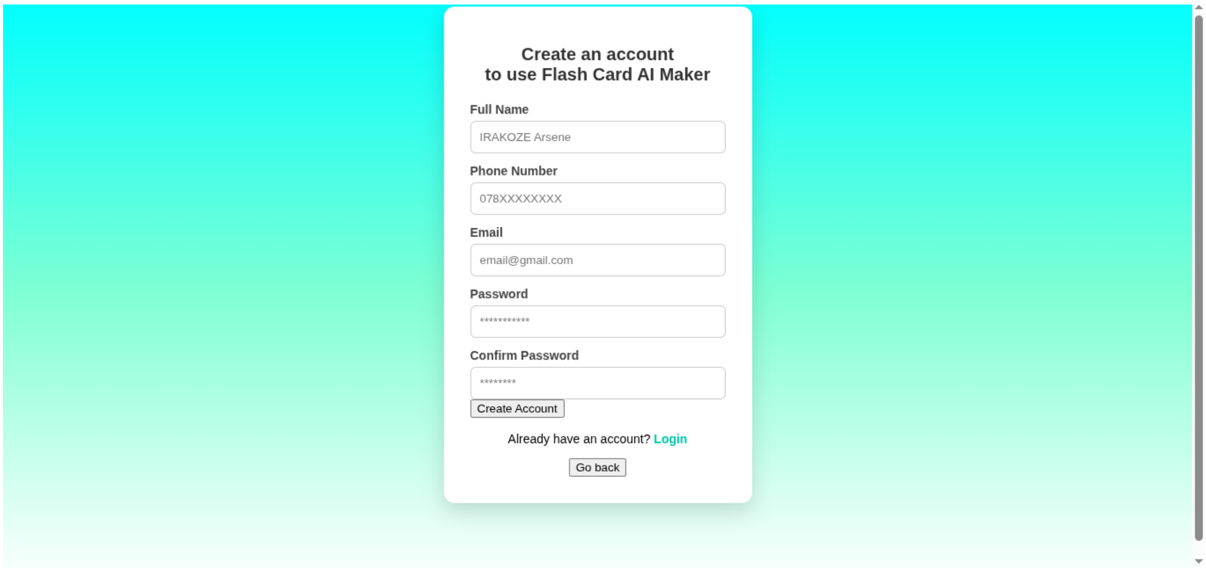
3.2.2 Login Page



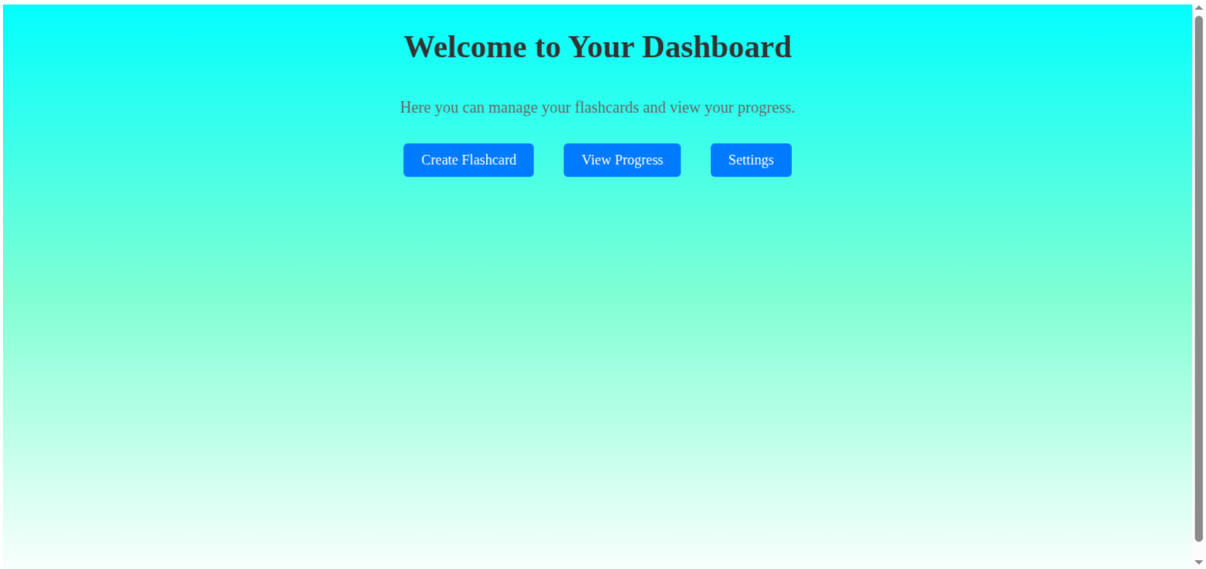
Discussion:

The login page allows authenticated access based on user roles.

3.2.3 Registration Page



3.2.4 Student Dashboard



3.2.5 Teacher Dashboard

Coming soon

3.2.6 Class Management Page

Coming soon

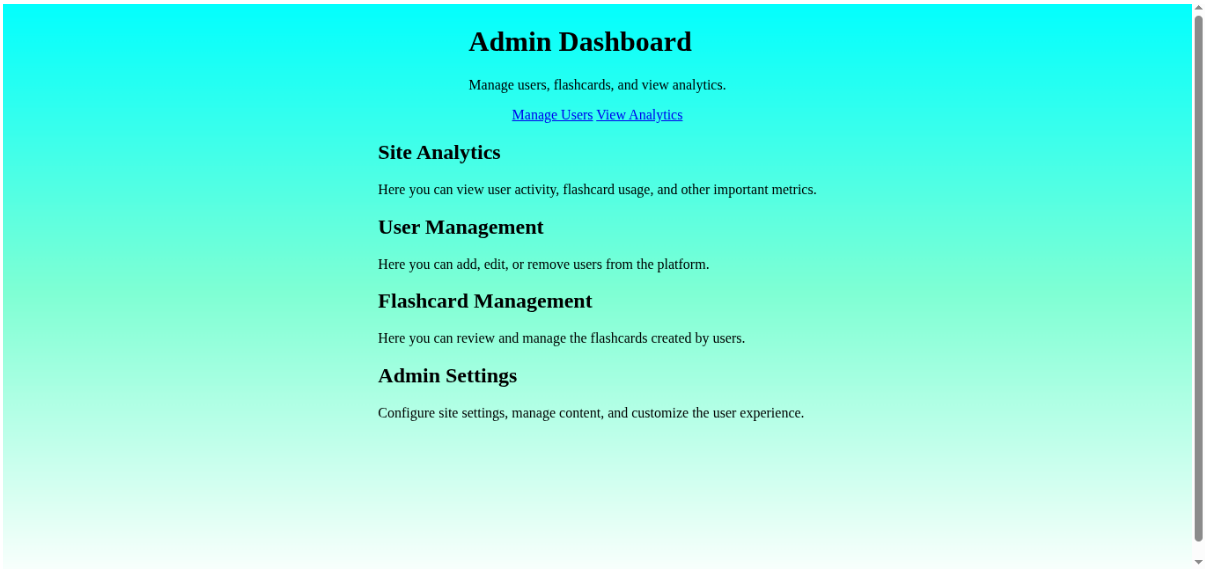
3.2.7 Flashcard Generation Page

Coming soon

3.2.8 Subscription Page

Coming soon

3.2.9 Admin Dashboard



Chapter 4. Conclusion and Recommendation

Conclusion

The AI Flashcard Maker system successfully implements most of the planned objectives, including authentication, AI-powered flashcard generation, classroom management, subscriptions, and administrative functionalities.

The system improves learning efficiency and provides an intelligent educational platform for both teachers and students.

Recommendations

To Lecturers

Provide additional guidance regarding software architecture and deployment practices.

To the School

Allocate more time for project development and testing activities.

To Educational Institutions

Adopt modern educational technologies that improve learning outcomes.

To Users

Provide training sessions to maximize system utilization.

To Future Researchers

Enhance the system by integrating mobile applications, advanced analytics, and adaptive learning techniques.

Appendix

GitHub Repository Link:

https://github.com/burnac321/AiFlashCardMaker.git